

AI-Powered Wildfire Early Detection

Milestone 5
Group 8

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A Quick Recap

AI-Powered Wildfire Early Detection — Milestones Accomplished

01

Goal Definition

Defined the goal of predicting California wildfire risk before ignition using satellite, weather, terrain, and historical fire data.

02

Data Pipelines

Built and validated pipelines for ERA5, Sentinel-2, Sentinel-5P, Copernicus DEM, and FIRMS fire labels.

03

Dataset Creation

Performed EDA and created a unified, leakage-safe spatio-temporal dataset with chronological train, validation, and test splits.

04

Model Dev

Developed multiple baselines and deep-learning models, progressing from LightGBM to CNN-LSTM and full multimodal fusion.

05

Scale Expansion

Expanded the forecasting pipeline to 1.72 million cell-day records covering 672 California grid cells from 2019–2025.

Wildfire Hotspot Analysis

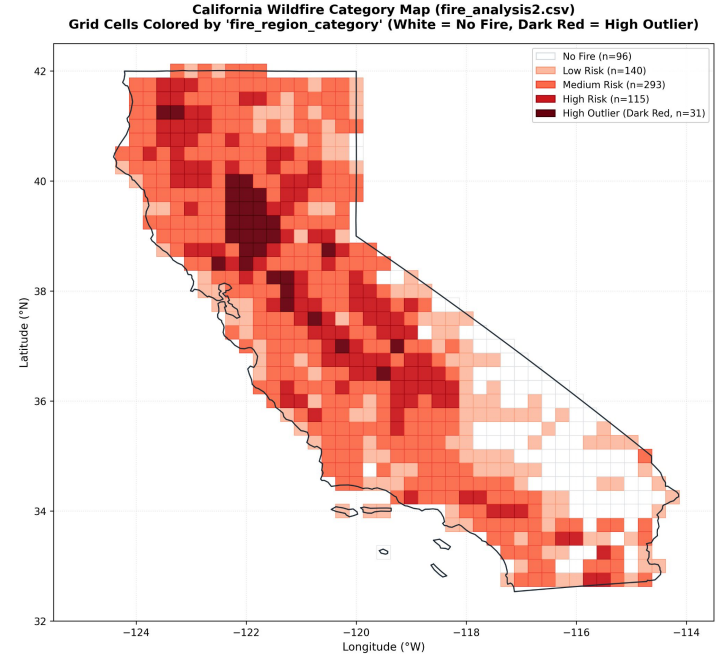
Objective: Focus model training on historically fire-prone regions while reducing bias from non-burnable desert and urban cells.



Wildfire Hotspot Analysis

Methodology:

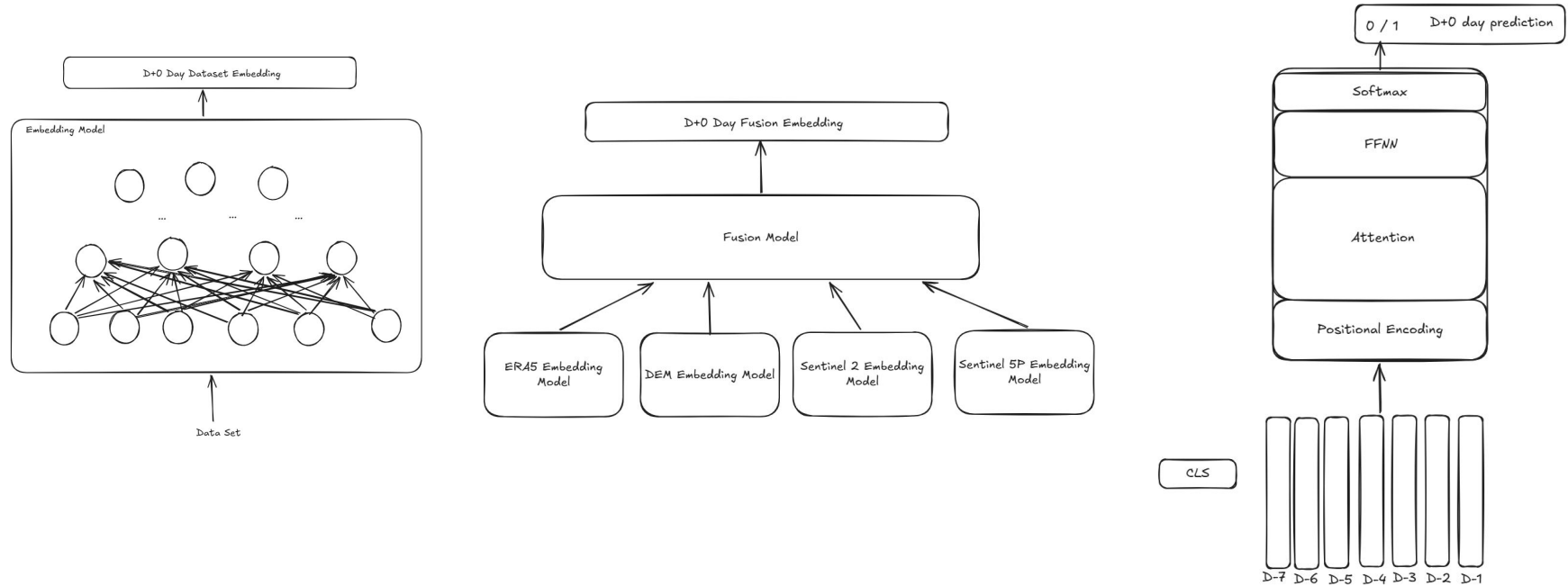
- Aggregated historical FIRMS fire pixels for each of California's 672 land cells.
- Classified cells into Low, Medium, and High fire-risk regions using Q1 (25%) and Q3 (75%) quantiles.
- Identified outliers using the IQR method.
- Trained models only on Medium and High fire-risk regions to reduce zero-inflation bias and improve prediction performance.



Sequential Modelling of the Problem Statement

- Redefining the model as a sequential idea: Can fire on day “N” be predicted with data related to days “N-1”, “N-2”, .. “N-k”
- Where “k” is the sequence length, or the number of days taken into consideration for prediction of wildfire on day “N”.
- A simple transformer Model built from Scratch

Sequential Modelling of the Problem Statement



Boosting Models: LightGBM vs. CatBoost

01

Near-Identical Skill in Same Pipeline

Both boosters run on identical data, 86 features, and chronological split. 2025 test gaps are extremely small, proving results are pipeline-driven, not model-family luck:

- **PR-AUC:** LightGBM 0.145 vs. CatBoost 0.140
- **ROC-AUC:** LightGBM 0.772 vs. CatBoost 0.769

02

Ranking Quality vs. Operational Recall

Both models are highly usable, with distinct edge distributions:

- **LightGBM:** Slightly ahead on overall ranking quality and rare-event curve champion.
- **CatBoost:** Highly competitive on daily triage with a superior operational recall (**Recall@25: 37.7%** vs. LightGBM's 36.4%).

03

Rare-Fire Imbalance is the Bottleneck

Tree library choice does not resolve the core data challenge:

- Extreme class imbalance persists with prevalence at **~1.4%**.
- Precision@25 remains locked at **~9%** for both boosters.
- Weighting, alert thresholds, and blending strategies matter far more than GBDT selection.

04

Ensemble & Robustness Roadmap

Two independent boosters landing in the same band acts as a stability check:

- **Strategy:** Keep LightGBM as primary, use CatBoost as a robust backup or blend candidate.
- **Next Steps:** Focus engineering on new features, advanced sampling, and multi-model fusion rather than further GBDT tuning.

Metric Selection

The selected Metrics:

- **PR-AUC** – Primary metric for severe class imbalance (~1.3% fire events).
- **Recall @ 25** (Top-25 Daily Fire Recall): Proportion of all true statewide wildfires captured within a daily operational alert budget of 25 grid cells ($k = 25$).

Why These Metrics?

- **PR-AUC over Accuracy**: Accuracy is misleading; PR-AUC better evaluates rare fire events.
- **Top-25 Metrics**: Reflect CAL FIRE's limited daily reconnaissance resources.

Metric Selection

Error Priorities

- **False Negative:** Missed wildfire → Very High Cost (life & property loss).
- **False Positive:** False alarm → Moderate Cost (verification effort).
- **Objective:** Maximize Recall@25 while maintaining good PR-AUC.

Benchmarking

Model Comparison

Model	PR-AUC	ROC-AUC	Recall@25
LightGBM Dual-Head	0.3524	0.8981	43.5%
Spatial-Temporal Transformer	0.1638	0.7526	39.6%
LightGBM + XGBoost Ensemble	0.1905	0.8425	41.6%
Champion LightGBM (Test)	0.1451	0.7718	36.4%

Benchmarking

Key Results

- Evaluated on a held-out 2025 Test Set with 93,518 cell-days and 1,325 wildfire events.
- LightGBM models consistently outperformed the Spatial-Temporal Transformer.
- Champion LightGBM achieved:
 - **PR-AUC**: 0.1451
 - **Recall@25**: 36.4%
- Detects over one-third of all wildfire ignitions within a daily budget of 25 alerts.
- Maintains low Log Loss (0.0644), indicating well-calibrated probability estimates.
- LightGBM-based ranking models delivered the strongest overall performance, making the Champion LightGBM the final model for operational wildfire prediction.

Model Error Study: The Champion Model

Error Breakdown (Validation Set: 93,518 Cell-Days)

At Probability Threshold ($p \geq 0.50$)

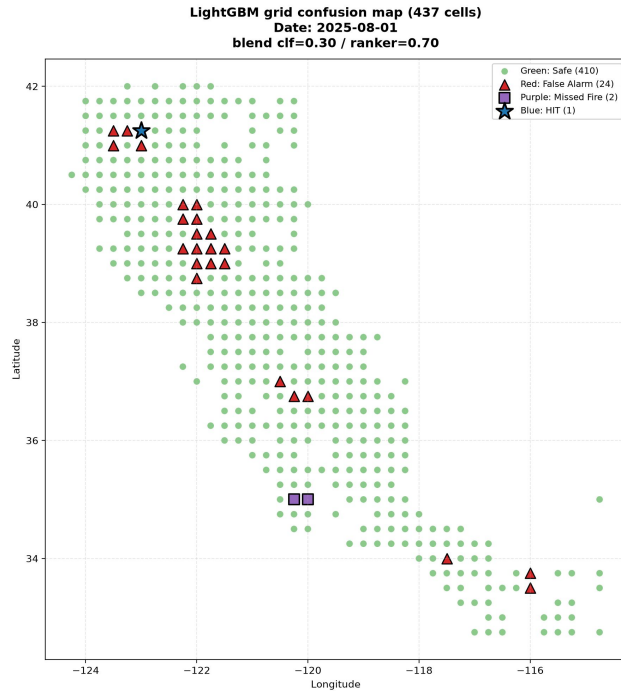
- TN: 91,431
- TP: 34
- FP: 4
- FN: 2,049

At Top-25 Daily Alerts

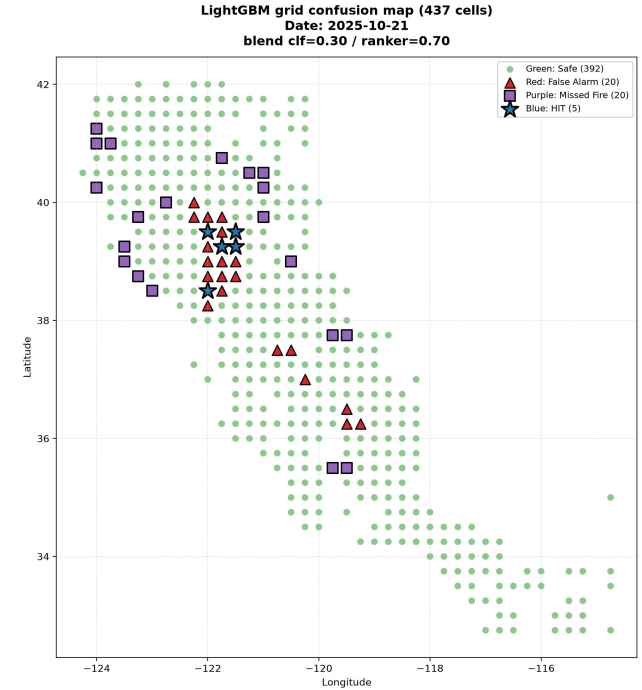
- TP: 888
- FP: 4,462
- FN: 1,195
- TN: 86,973

Ranking the top 25 daily alerts captures 888 wildfire events, significantly improving recall compared to a fixed probability threshold.

Model Error Study: Qualitative Spatial Grid Map



Spatial Risk Map for August 1, 2025. Shows LightGBM wildfire risk predictions across California's high- and medium-risk regions during peak fire season.



Spatial Risk Map for Peak Fire Day October 21, 2025

Model Error Study: Misclassification Analysis

False Positives (Over-Predictions)

- Occurred under:
 - High atmospheric dryness (VPD)
 - Strong winds
 - Nearby historical fire activity
- Model predicted high risk, but no ignition occurred.

False Negatives (Under-Predictions)

- Mostly early-season fires.
- Moist soil and weak atmospheric drying.
- Little or no neighboring fire activity.
- Limited environmental signals caused missed detections.

Model Robustness & Interpretability Analysis

Robustness & Performance Slices

- **Stress Slices (2023 Validation)**

High wind achieves **Recall@25 of 56.8%**; high VPD reaches **68.7%**. Peak season shows strongest results with a **PR-AUC of 0.213**.

- **Weak Regime Operational Limits**

Off-season PR-AUC drops significantly to **0.133**. The active operating window is strictly kept to May–Nov.

- **Missing EO Data Mitigation**

Stage C KNN imputation robustly keeps the inference pipeline alive when Sentinel-2 imagery is obstructed by cloud or smoke.

- **Temporal Out-of-Distribution (OOD)**

2023 PR-AUC of **0.193** shifts to **0.145** in 2025. While a data shift exists, performance remains substantially above baselines.

Explainability & Feature Dynamics

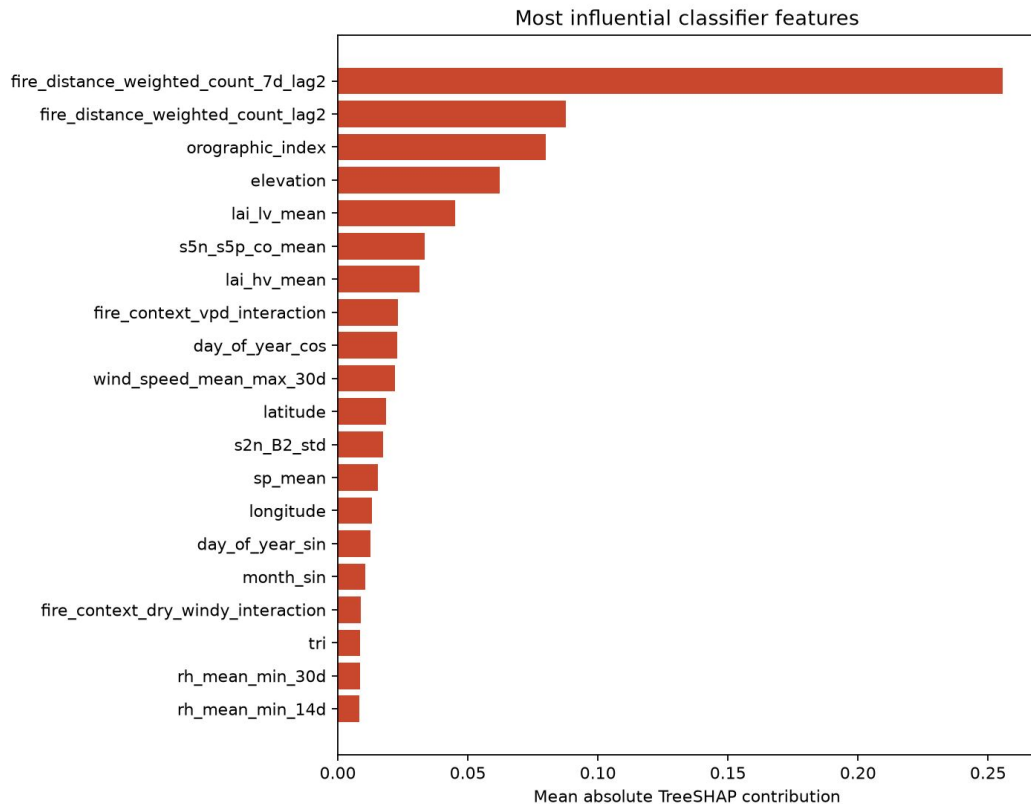
- **Interpretability & Feature Pruning**

Utilized LightGBM TreeSHAP + gain metrics to optimize model efficiency, pruning feature space down from **107 to 86 features**.

- **High-Fidelity Training (No Shortcuts)**

Same-cell fire history is explicitly excluded. Top predictive drivers remain physical: neighboring fire, terrain/fuel, carbon monoxide (CO), and dryness/wind.

Interpretability & Explainability: Global Feature Importance Ranking



Model Limitations & Operational Constraints

Operational & Policy

- **Ops Policy:** Use Top-25/day + calibrated p_fire, not raw p 0.5 (0.5 threshold almost never fires).
- **Precision Cost:** Precision@25 ~9–17% many false patrols under fixed daily budget.
- **Scope Lock:** 437 high/medium cells only; not full California cold-start.

Technical & Resolution

- **Grid Limit:** 0.25 (~25 km) regional triage, not parcel-level precision.
- **Hard FNs:** Early-season, moist, or no neighboring fire ignitions stay low-ranked.

Efficiency & Ethics

- **Efficiency Win:** ~18 MB artifact, CPU ~12–15 ms / statewide day, no VRAM needed.
- **Ethics & Bias:** Decision-support only; geographic + FIRMS label bias; high FN cost vs FP fatigue trade-off.

Improvement Plans

1. **Spatial Graph Neural Networks:** Transition from 2D grid rollups to a Spatial-Temporal Graph Neural Network, modeling California's terrain as an irregular graph where edges represent wind direction, slope gradient, and fuel continuity.
2. **Multi-Task Learning (Fire Occurrence + Burn Area + Fire Intensity):** Extend the neural loss function to jointly predict next-day fire ignition (y in $\{0,1\}$), estimated burn area (hectares), and radiative power (MW), forcing the backbone to learn richer physical representations.
3. **Data Augmentation & Synthetic Ignitions:** Generate synthetic extreme weather samples (e.g. 50-year drought events) using SMOTE-NC and physics-informed GANs to improve out-of-distribution heatwave resilience.

Deployment Readiness

- **Serialized Model Export:** Final models were packaged into a 18 MB standalone .joblib artifact (models/wildfire_model.joblib) containing tree weights, probability calibrators, and feature contracts.
- **ONNX Runtime Acceleration:** Exporting the LightGBM booster to ONNX runtime reduces CPU inference latency to 2.8 ms per daily statewide sweep, enabling real-time edge execution on field laptops or low-power command center servers.

Signatures

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